

ISE 323: Human-Computer Interaction

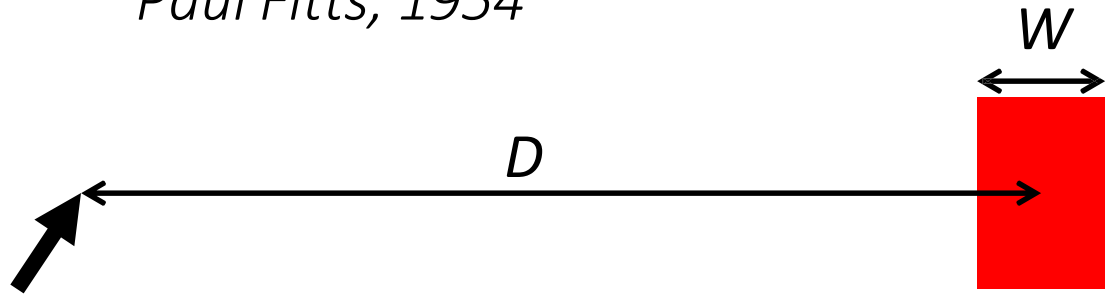
Trends and Practices in HCI SBIAHU

Shubham Jain
Stony Brook University

Review

Fitts' Law

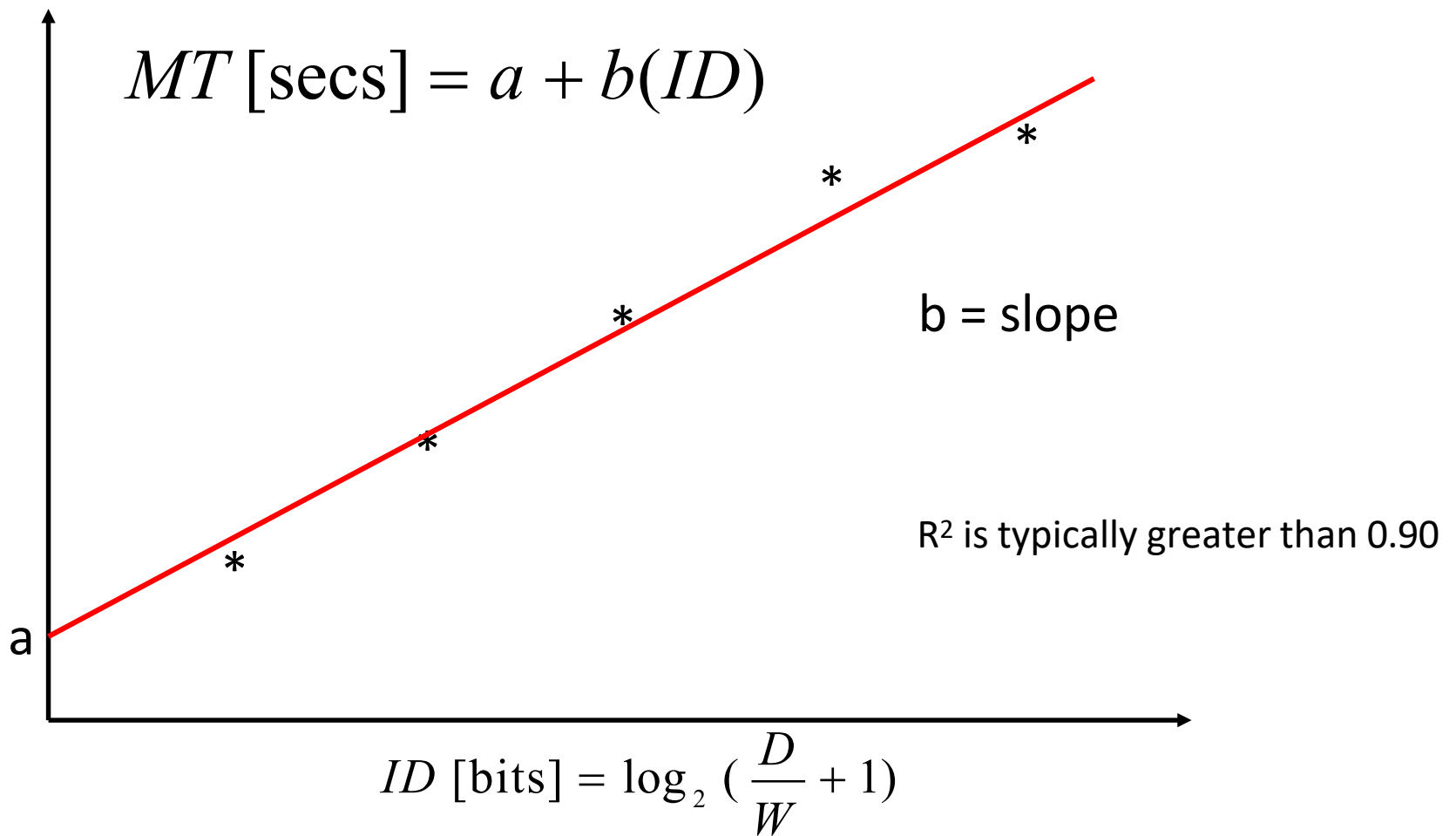
Paul Fitts, 1954



$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

Movement Time

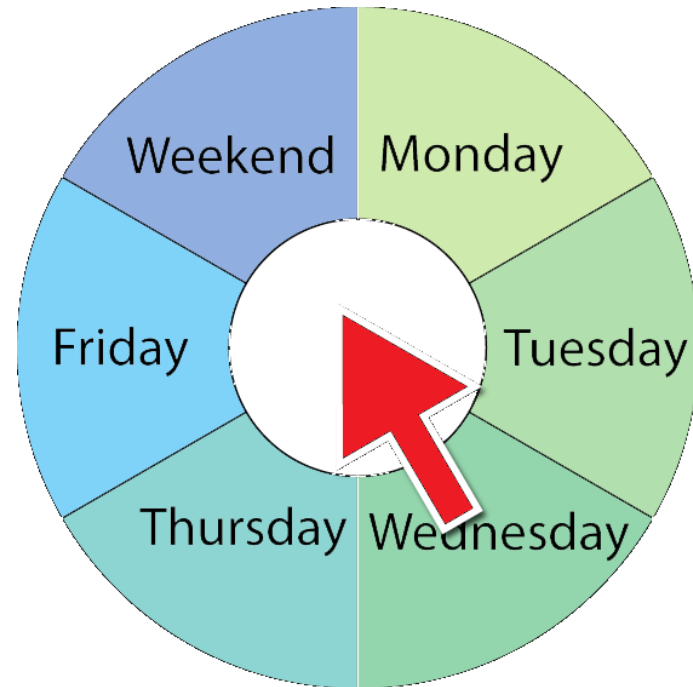
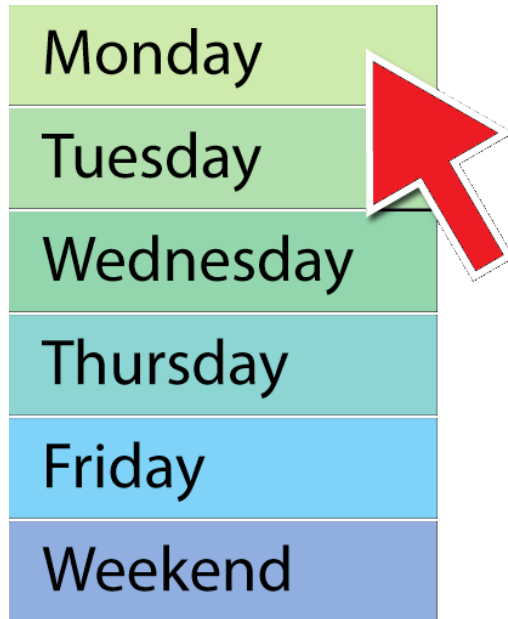
Index of Difficulty (ID)



$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

Only Relative Precision (D / W) Matters

Fitts' Law Example



Which will be faster on average?

pie menu (bigger targets & less distance)

Qwerty Layout



Q	W	E	R	T	Y	U	I	O	P
A	S	D	F	G	H	J	K	L	
	Z	X	C	V	B	N	M		

Qwerty is inefficient for one finger typing.

Soft keyboard

- Besides handwriting recognition, one popular method of text input for pen-based mobile systems is through a soft keyboard.
- Users enter text by tapping on the image of a keyboard on the system's display.
-
- Although the QWERTY layout is entrenched for physical keyboards, soft keyboards are easy to implement and modify.

Usability Issue

- A usability issue is learning time. Users who bring desktop computing experience to mobile computing may fare poorly on a non-QWERTY layout - at least initially.
- Thus, longitudinal empirical testing is important.
- We want to establish not only a layout's potential for experts, but also the learning time for typical users to meet and exceed entry rates with a QWERTY layout.

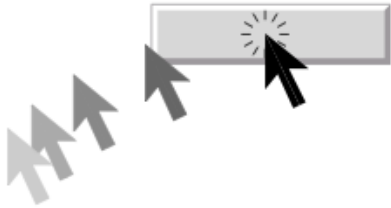
Quasi-Qwerty Layout

Q	W	D	R	T	U	Y	L	K	P
Z	A	S	E	H	N	I	O	M	
	X	F	V	C	G	B	J		

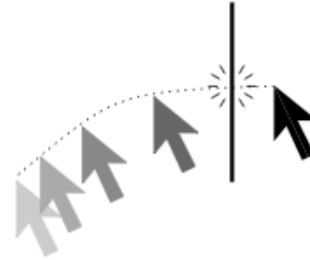
Qwerty

Q	W	E	R	T	Y	U	I	O	P
A	S	D	F	G	H	J	K	L	
	Z	X	C	V	B	N	M		

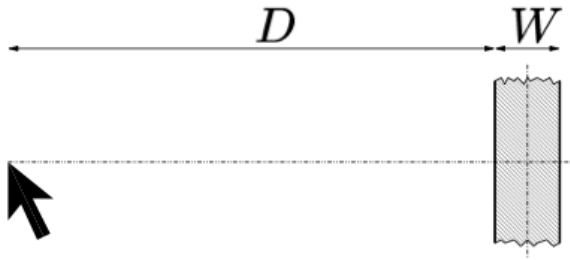
Pointing vs Crossing



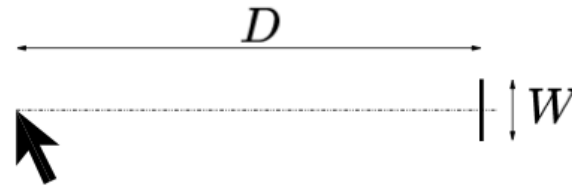
(a) Pointing a target



(b) Crossing a goal



(a) Pointing: variability is allowed in the direction collinear to movement



(b) Crossing: variability is allowed in the direction orthogonal to movement

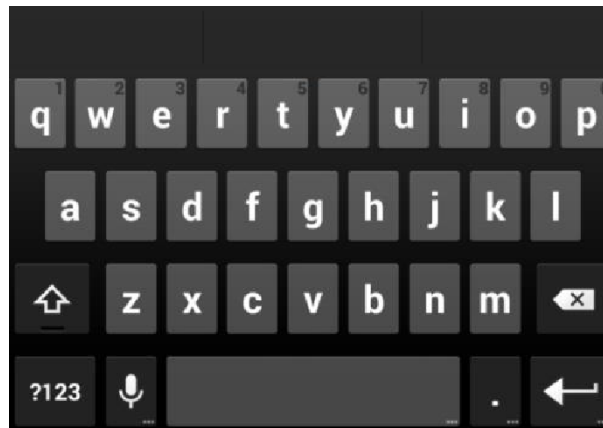
Moving on

Smart Touch Keyboard

Typed Word

Keyboard Output

agsim



again

quivj



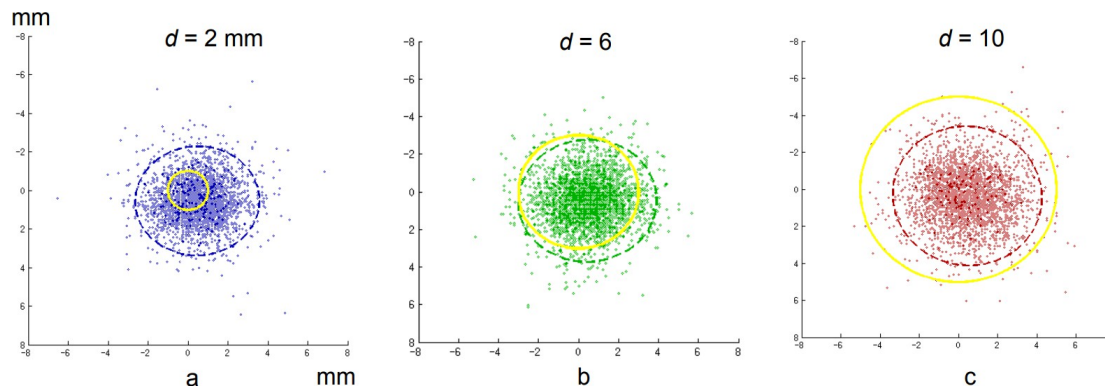
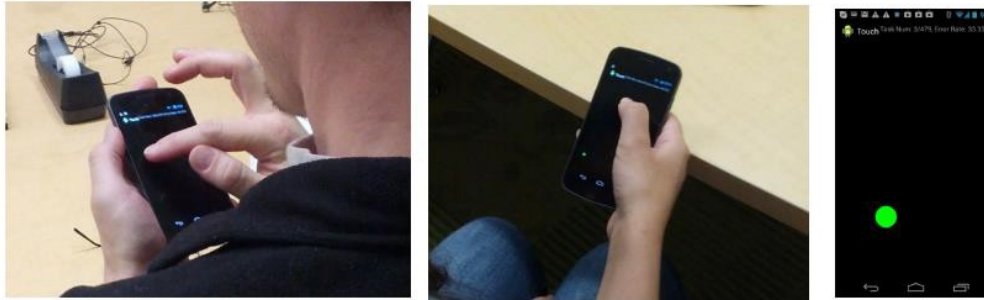
quick

fav

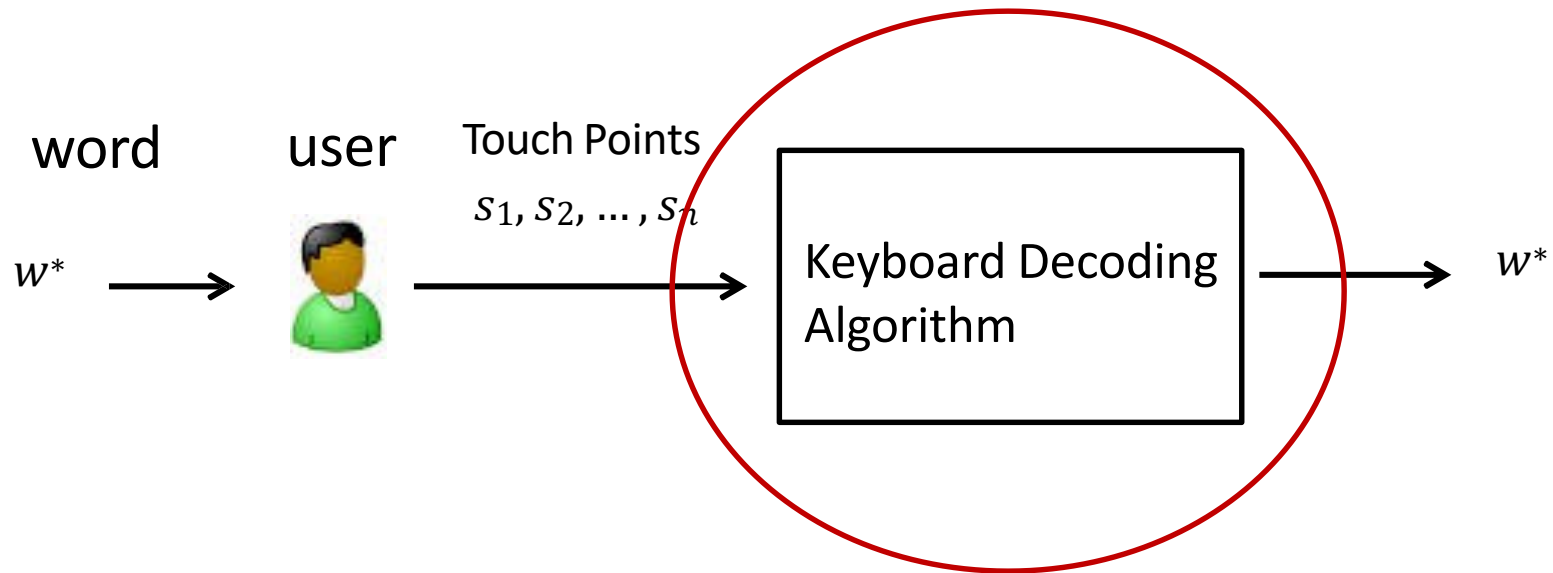


favorite

Understanding Touch



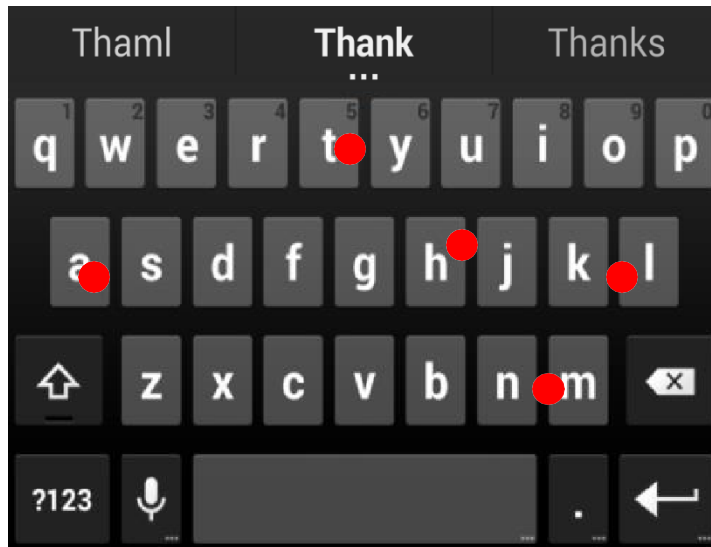
Evaluation of Decoding Algorithm



Modern Touchscreen Keyboard

Correction

Thaml → Thank

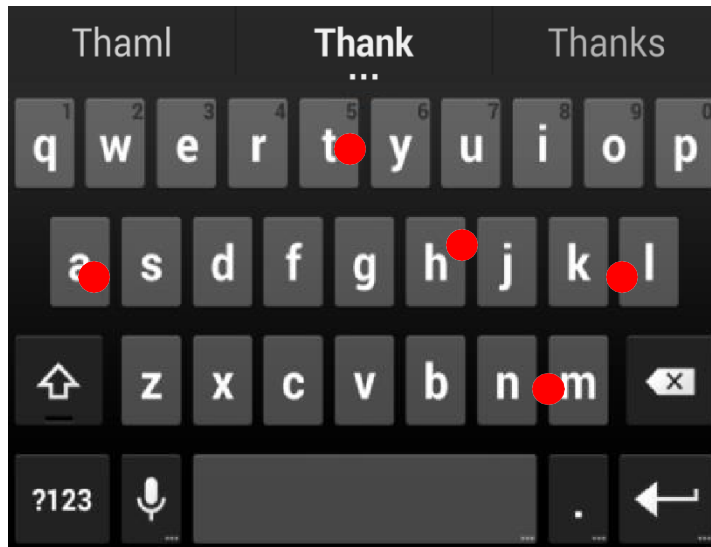


$$Success\ Rate(W) = \frac{Correct\ Words}{Total\ Words}$$

Modern Touchscreen Keyboard

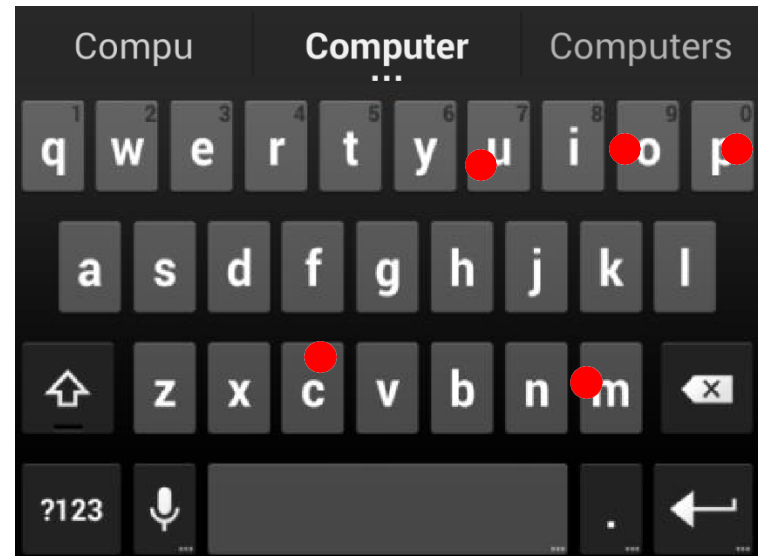
Correction

Thaml → Thank



Completion

Compu → Computer

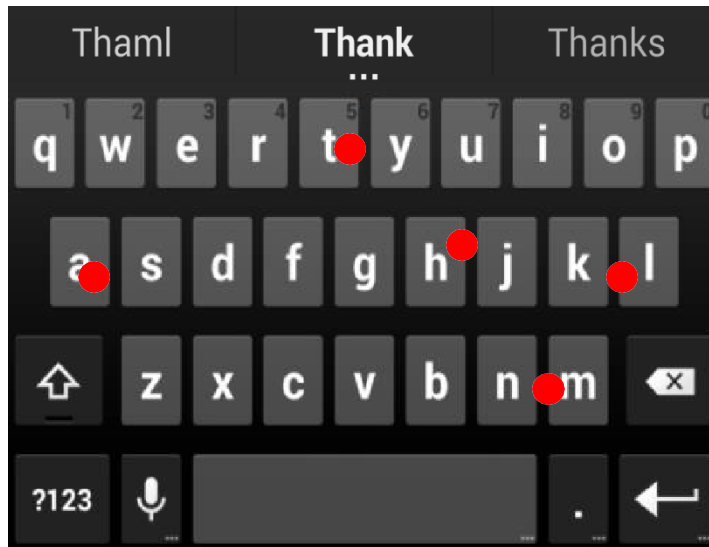


$$\text{Success Rate}(W) = \frac{\text{Correct Words}}{\text{Total Words}}$$

Modern Touchscreen Keyboard

Correction

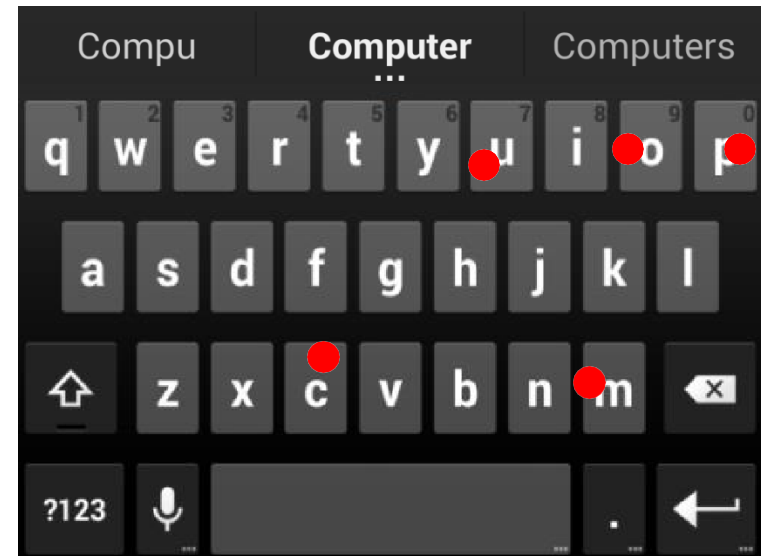
Thaml → Thank



$$\text{Success Rate}(W) = \frac{\text{Correct Words}}{\text{Total Words}}$$

Completion

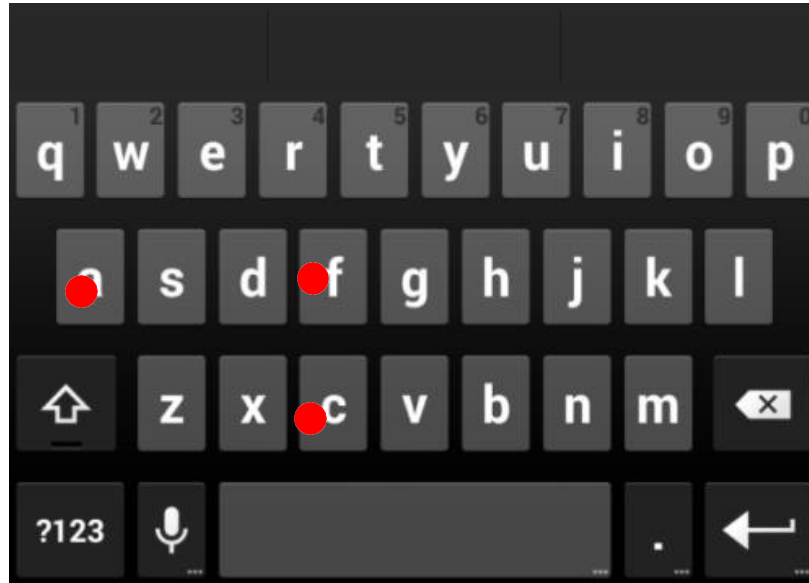
Compu → Computer



$$\text{Keystroke Saving}(S) = \frac{\text{Saved Keystrokes}}{\text{Maximum Keystrokes}}$$

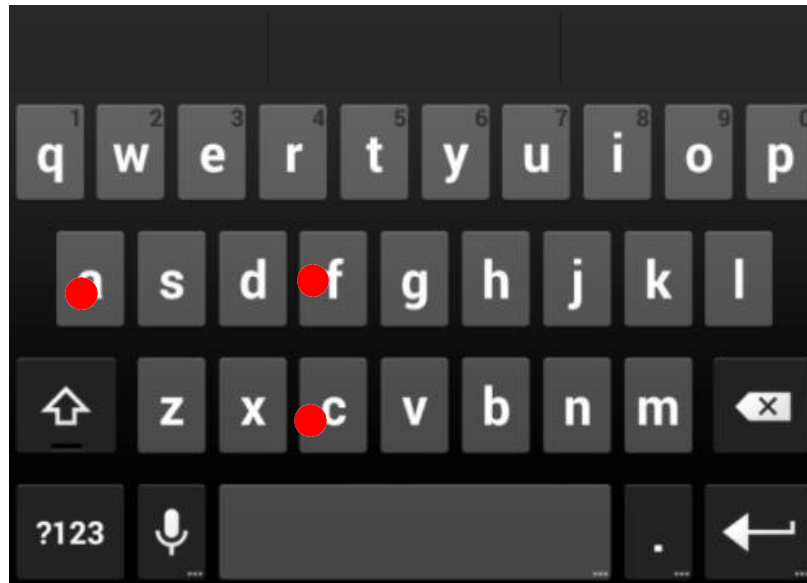
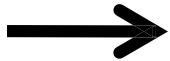
Correction vs. Completion

fac →



Correction vs. Completion

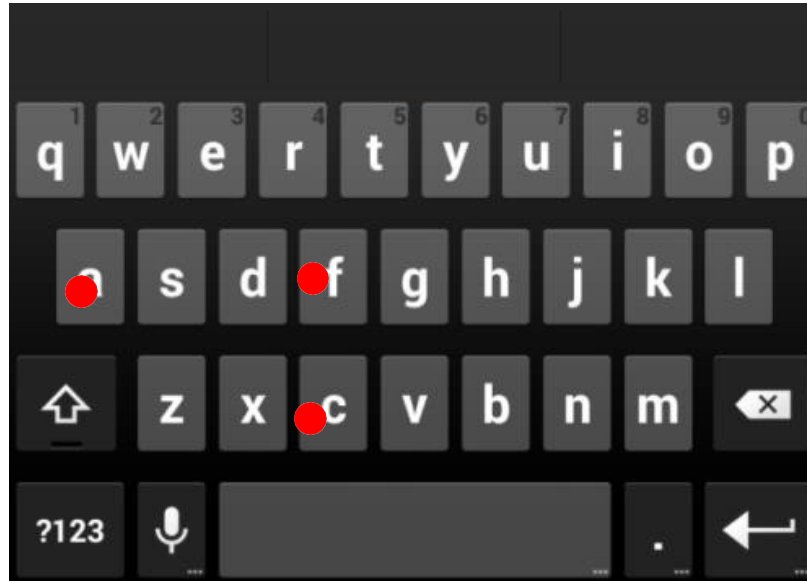
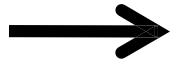
fac



fax

Correction vs. Completion

fac

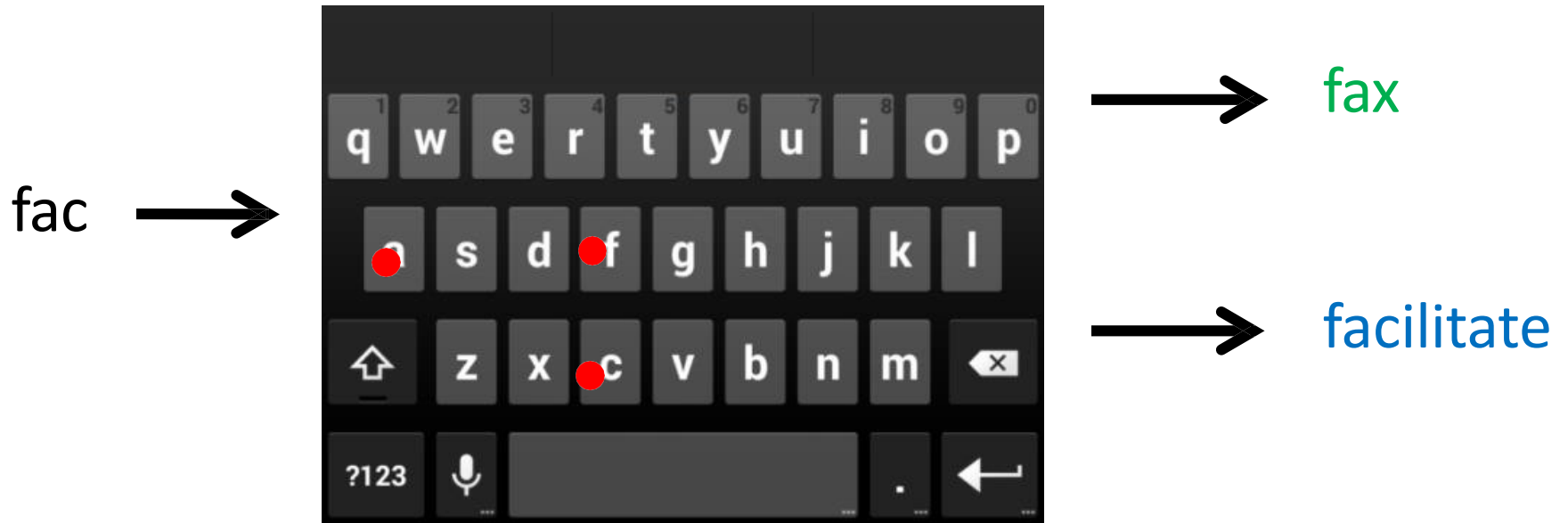


fax



facilitate

Correction vs. Completion



Relationship between Correction and Completion ?

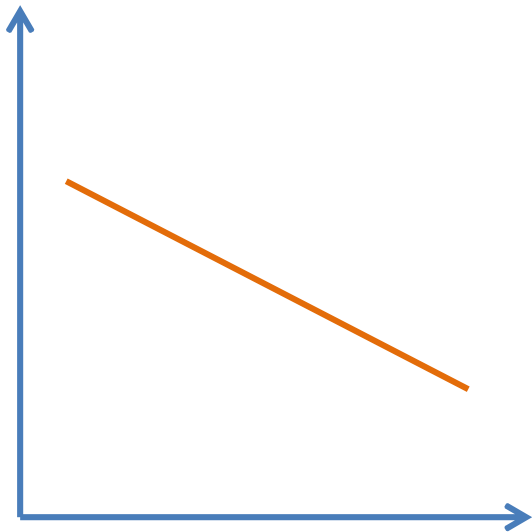
Correction vs. Completion

3 Possible Relationships

Correction vs. Completion

3 Possible Relationships

Completion



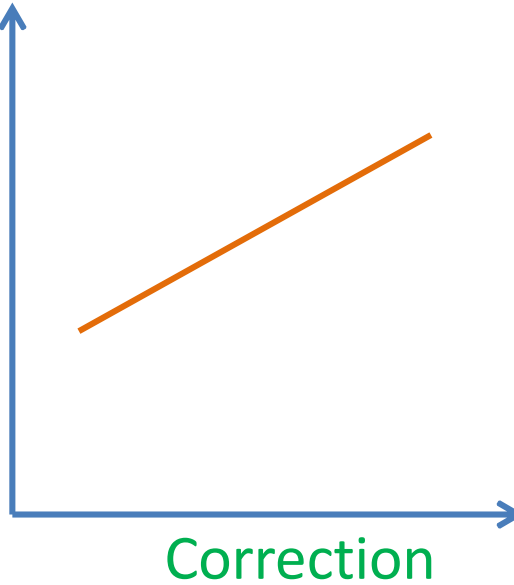
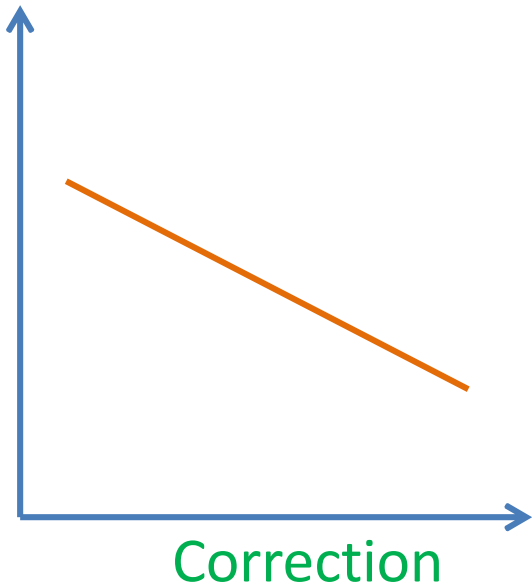
Correction

Correction vs. Completion

3 Possible Relationships

Completion

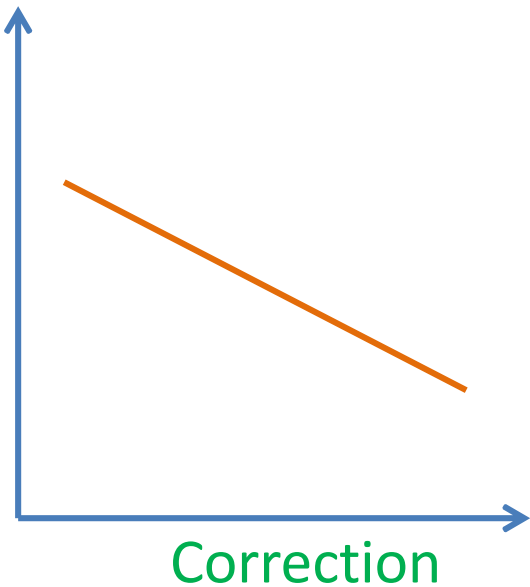
Completion



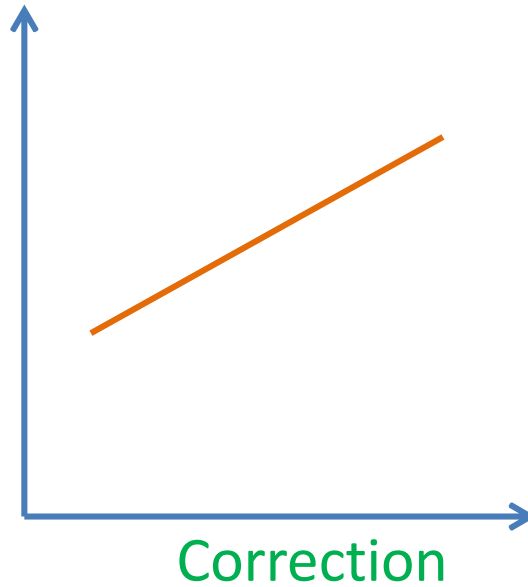
Correction vs. Completion

3 Possible Relationships

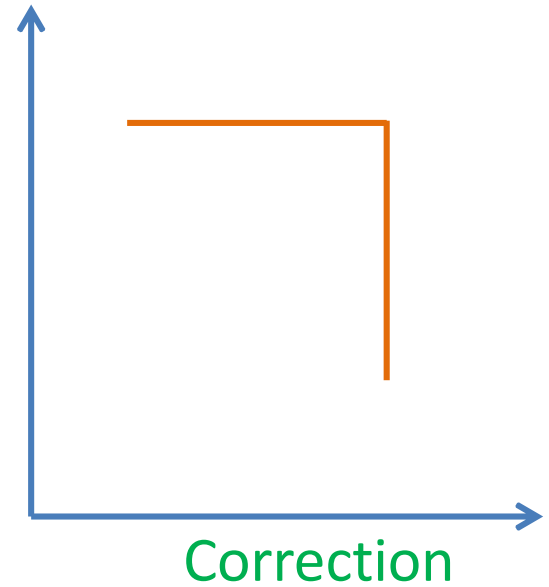
Completion



Completion



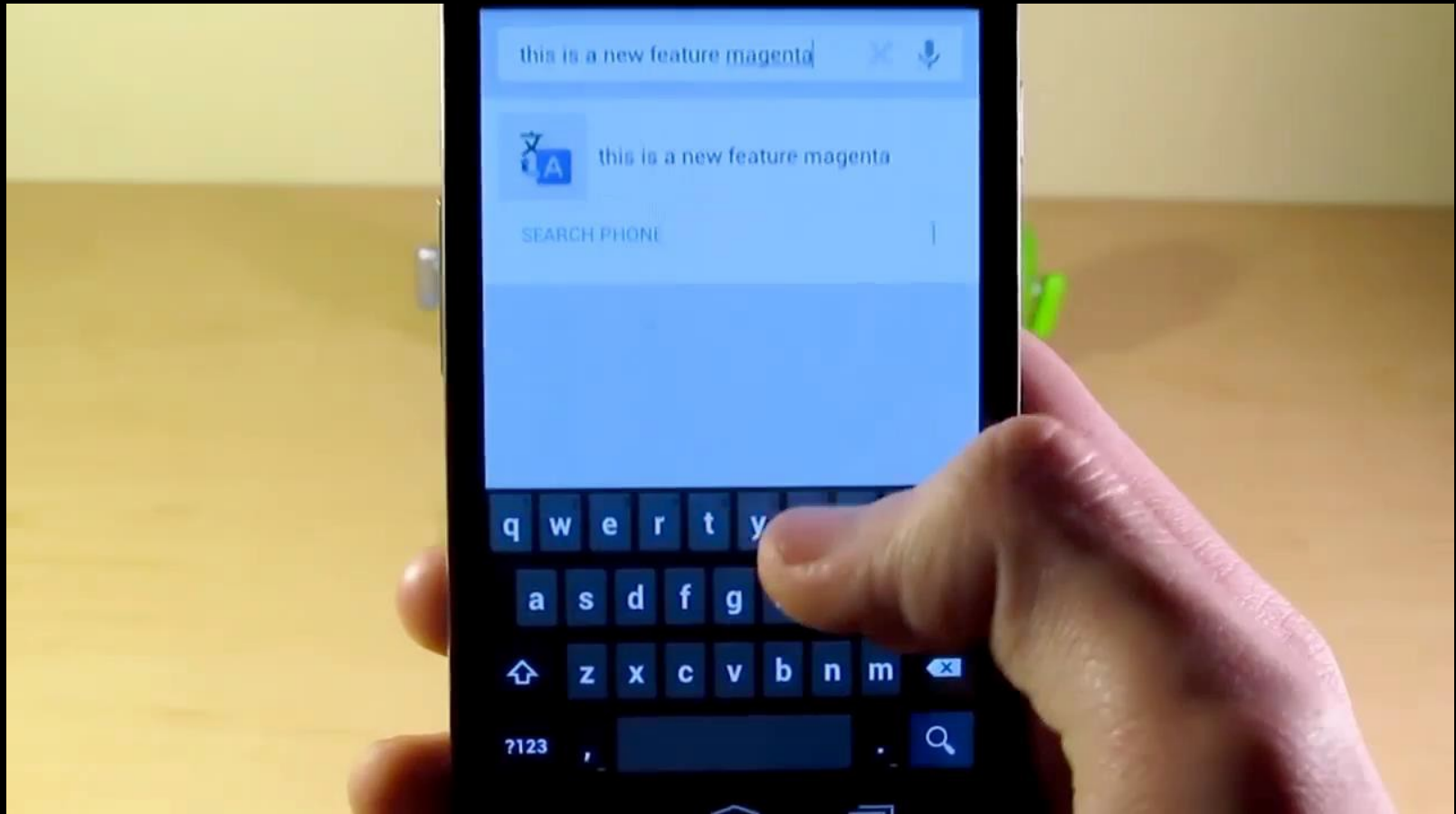
Completion



Keyboard Layout Optimization for Gesture Typing

Gesture Typing: Overview

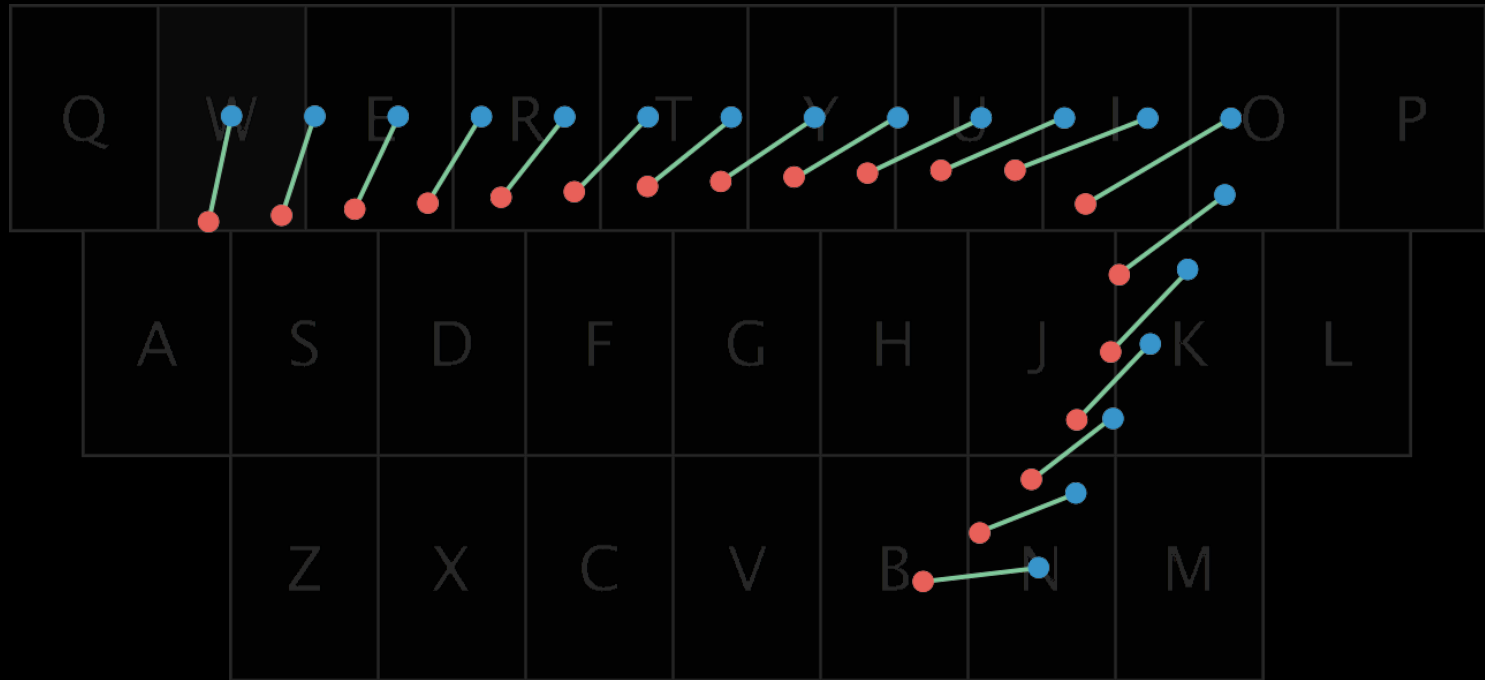
Gesture Typing: Typing words by swiping through their letters



Proportional Shape Matching



Proportional Shape Matching



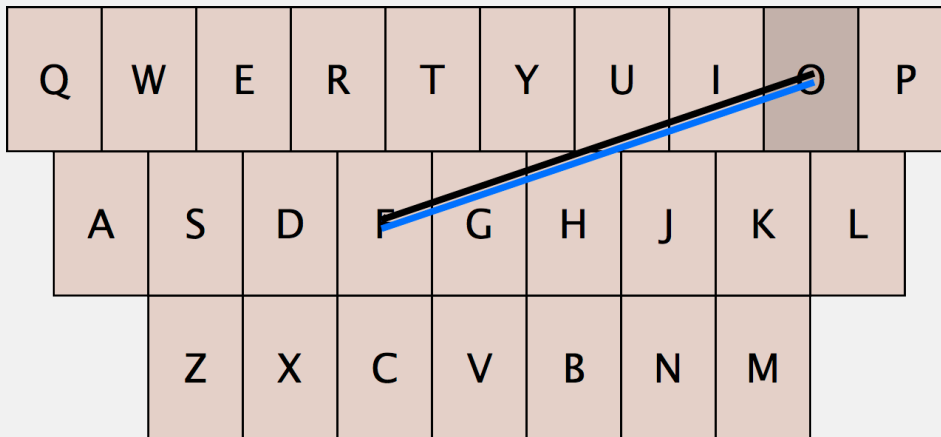
$$d(w, x) = \frac{1}{N} \sum_{i=1}^N \|w_i - x_i\|_2$$

Key Disadvantage:
Gesture Ambiguity

Question 1

of

off



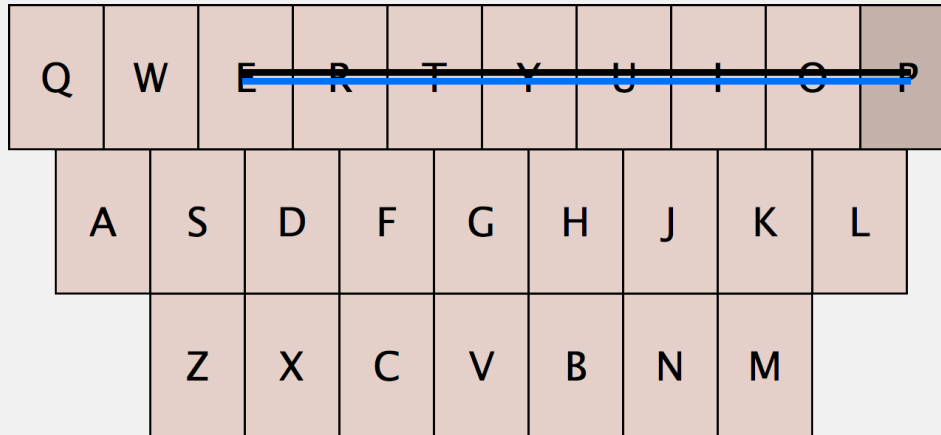
Answer:

SAME

Question 2

pretty

prey



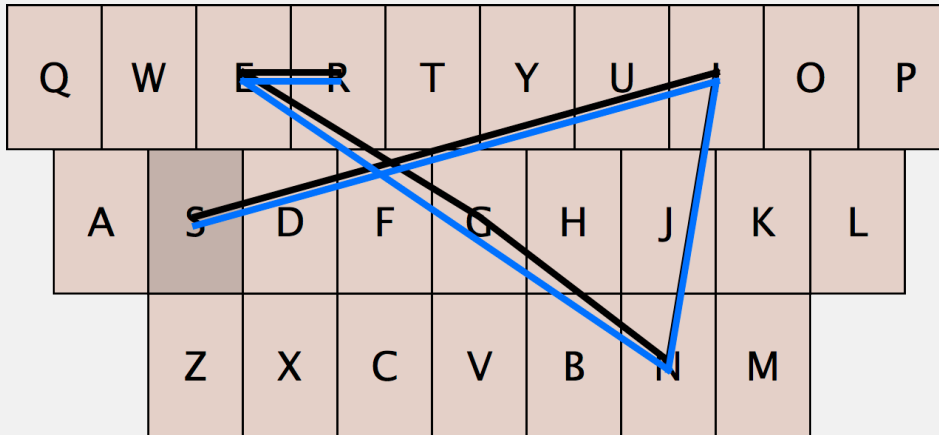
Answer:

SAME

Question 3

singer

sinner



Answer:

DIFFERENT

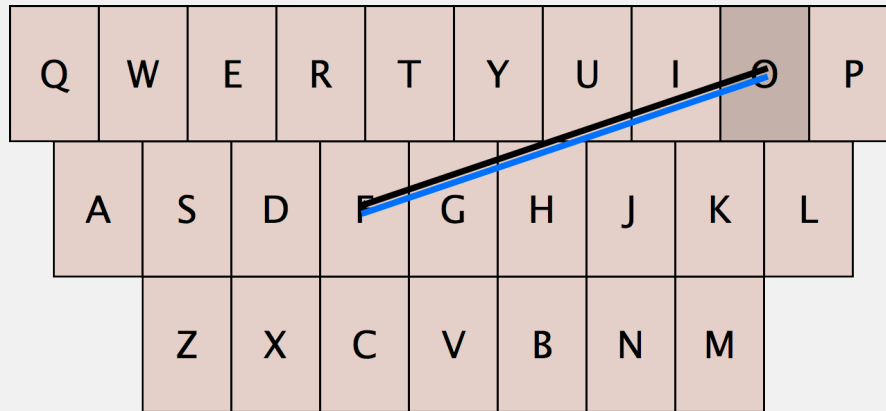
Gesture Ambiguity

Types of Ambiguity

Question 1

of

off

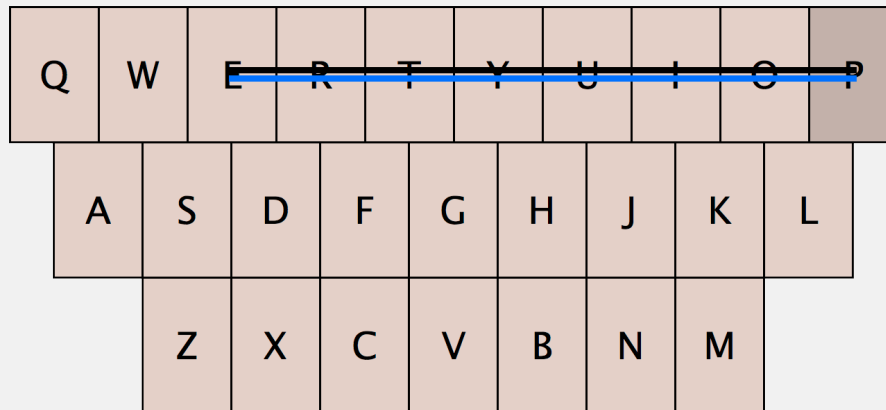


Types of Ambiguity

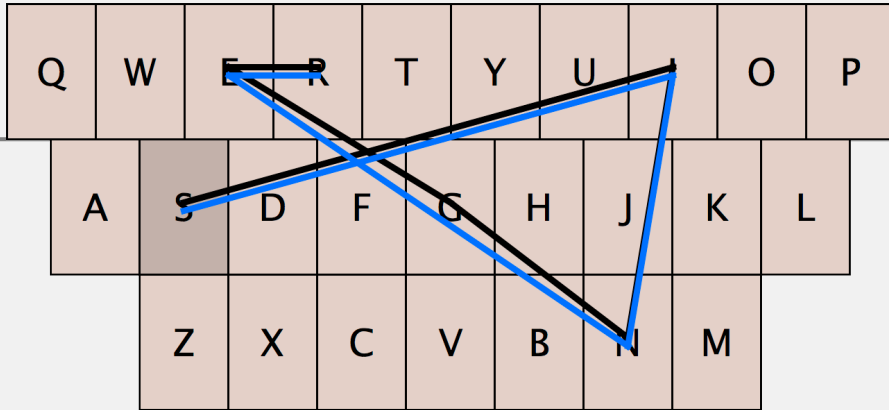
Question 2

pretty

prey



Types of Ambiguity

Type		% of Words
Ide (Unav)	Question 3 sin ger sinner	.2%
Ide (Avo)		.7%
Sin (Avo)		

Types of Ambiguity

Type	Examples	% of Words	
		by Count	by Frequency
Identical (Unavoidable)	of off	4.1%	22.2%
Identical (Avoidable)	pretty prey	2.3%	5.7%
Similar (Avoidable)	singer sinner	77.8%	69.0%

Only 3.1% of words have distinct gestures!

Optimized Layouts

R	Q	A	J	T	U	Y	I	P	H
W	O	Z	V	F	X	K	G	N	
	E	B	S	M	D	C	L		

GK-C

K	G	D	N	I	L	Y	B	J	Z
V	C	A	E	R	O	M	F	Q	
	W	H	T	S	U	P	X		

GK-S

B	Q	P	C	R	J	E	V	D	L
M	Y	Z	S	I	T	H	X	K	
	F	U	W	O	N	A	G		

GK-D

Q	D	W	S	O	I	Y	U	J	P
Z	R	F	A	T	N	G	K	L	
	C	E	X	H	V	M	B		

GK-T

Other principles for layout optimization

Gestalt Principles

- Gestalt Principles are principles/laws of human perception that describe how humans group similar elements, recognize patterns and simplify complex images when we perceive objects.
- Designers use the principles to organize content on websites and other interfaces so it is aesthetically pleasing and easy to understand.

Gestalt Principles

Based on psychology of how humans **perceive patterns** and **organize visual elements**

Create designs that are:

- more intuitive
- visually pleasing and organised
- guide users focus
- easy to navigate

Gestalt Principles

- "Gestalt" is German for "unified whole".
- German psychologists Max Wertheimer, Kurt Koffka, and Wolfgang Kohler created the Gestalt Principles in the 1920s.
- They wanted to understand how people make sense of the confusing things they see and hear.
- They identified a set of laws that address the natural compulsion to find order in disorder.
- According to this, the mind "informs" what the eye sees by perceiving a series of individual elements as a whole.
- Graphic designers quickly embraced Gestalt Principles, using them to create eye-catching designs with well-placed elements.

Gestalt Laws

- Law of Similarity
- Law of Continuation
- Law of Closure
- Law of Symmetry
- Law of Proximity
- Law of Focal Point

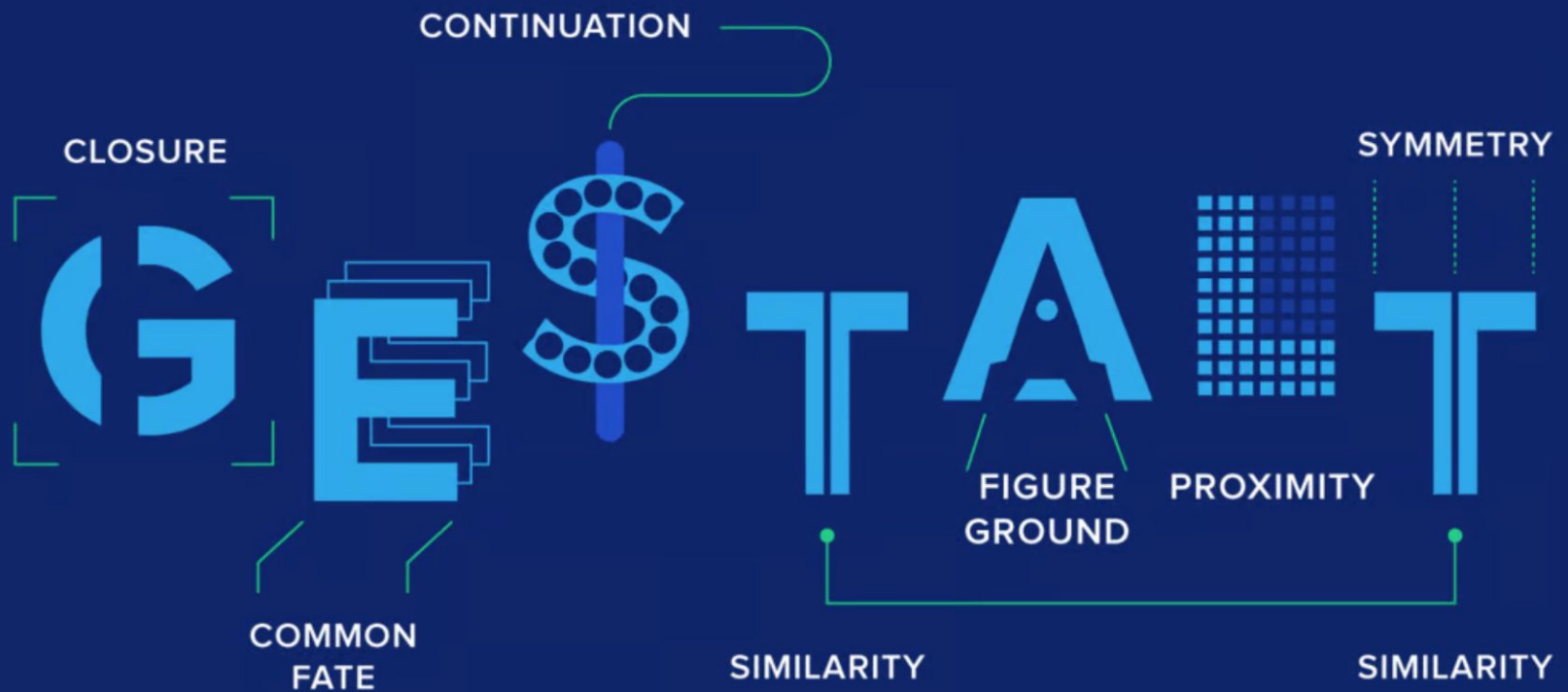


Image from <https://www.toptal.com/designers/ui/gestalt-principles-of-design>. All rights reserved.

CONTINUATION

Closure: Our minds tend to complete shapes even when parts are missing, which is often used in logos and icons to suggest forms without fully defining them.

CLOSURE

CLOSURE

SYMMETRY

COMMON FATE

SIMILARITY

SIMILARITY

FIGURE GROUND

PROXIMITY

Image from <https://www.toptal.com/designers/ui/gestalt-principles-of-design>. All rights reserved.

Closure

- We prefer complete shapes, so we automatically fill the gaps between elements to perceive a complete image.
- That's how we can see the whole first. You can use closure creatively to gain users' trust and admiration.
- Users will appreciate it when they see pleasing "wholes" made from cleverly placed elements like lines, dots, or shapes.

Closure



Iconic logos like IBM's and the World Wildlife Fund's are great examples of closure. IBM's logo has blue lines in three stacks. WWF's logo has black shapes on a white background that we interpret as the shape of a panda.

CONTINUATION

Common Fate: Elements moving in the same direction are perceived as related. This principle can apply to animations or grouped elements with similar directional cues, helping users understand which items belong together.

RY

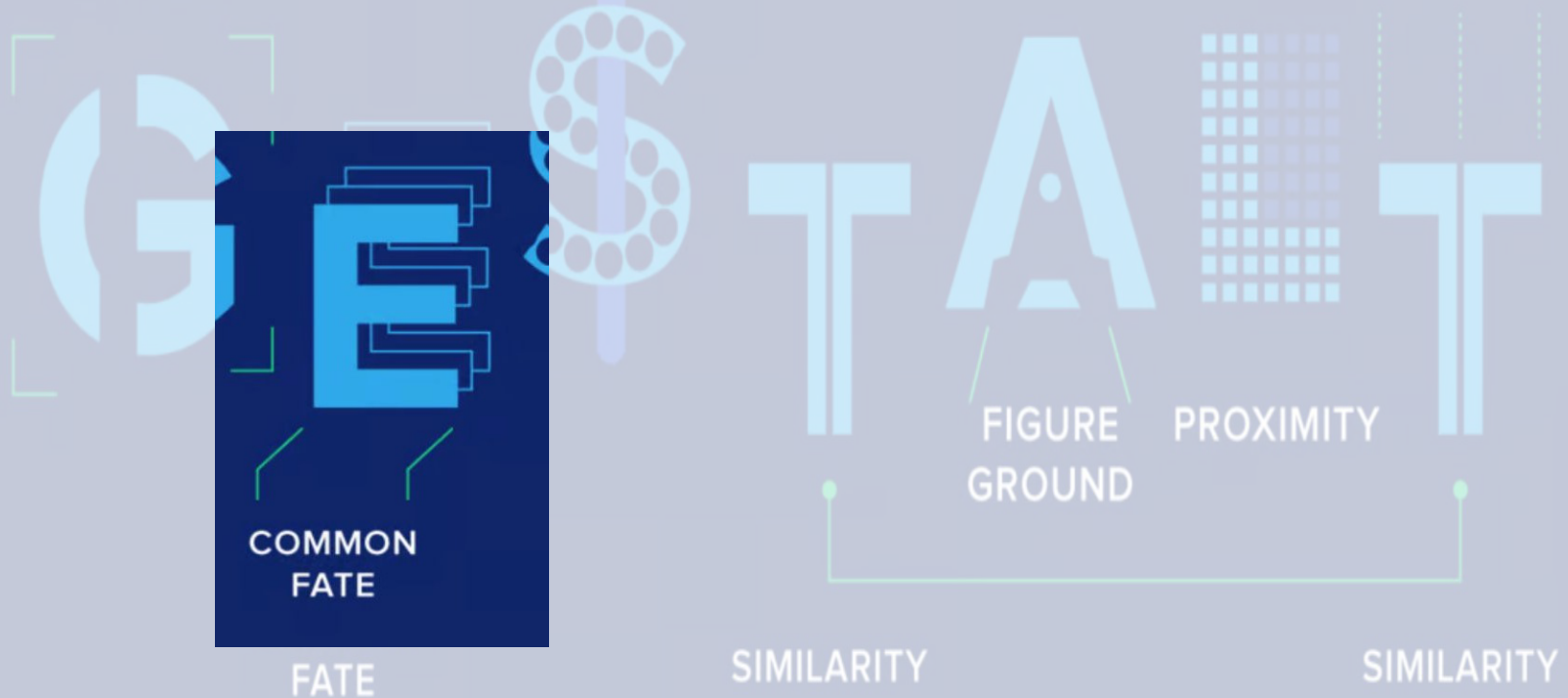
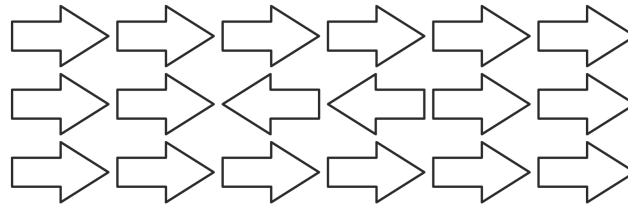


Image from <https://www.toptal.com/designers/ui/gestalt-principles-of-design>. All rights reserved.

Common Fate

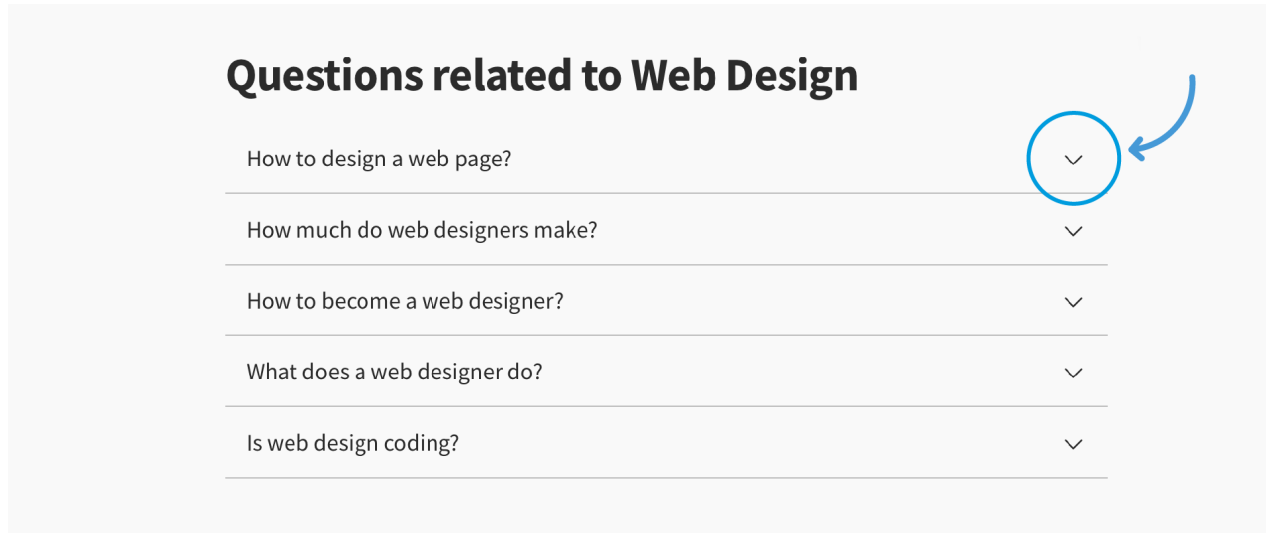
Gestalt Rule: **Common Fate**



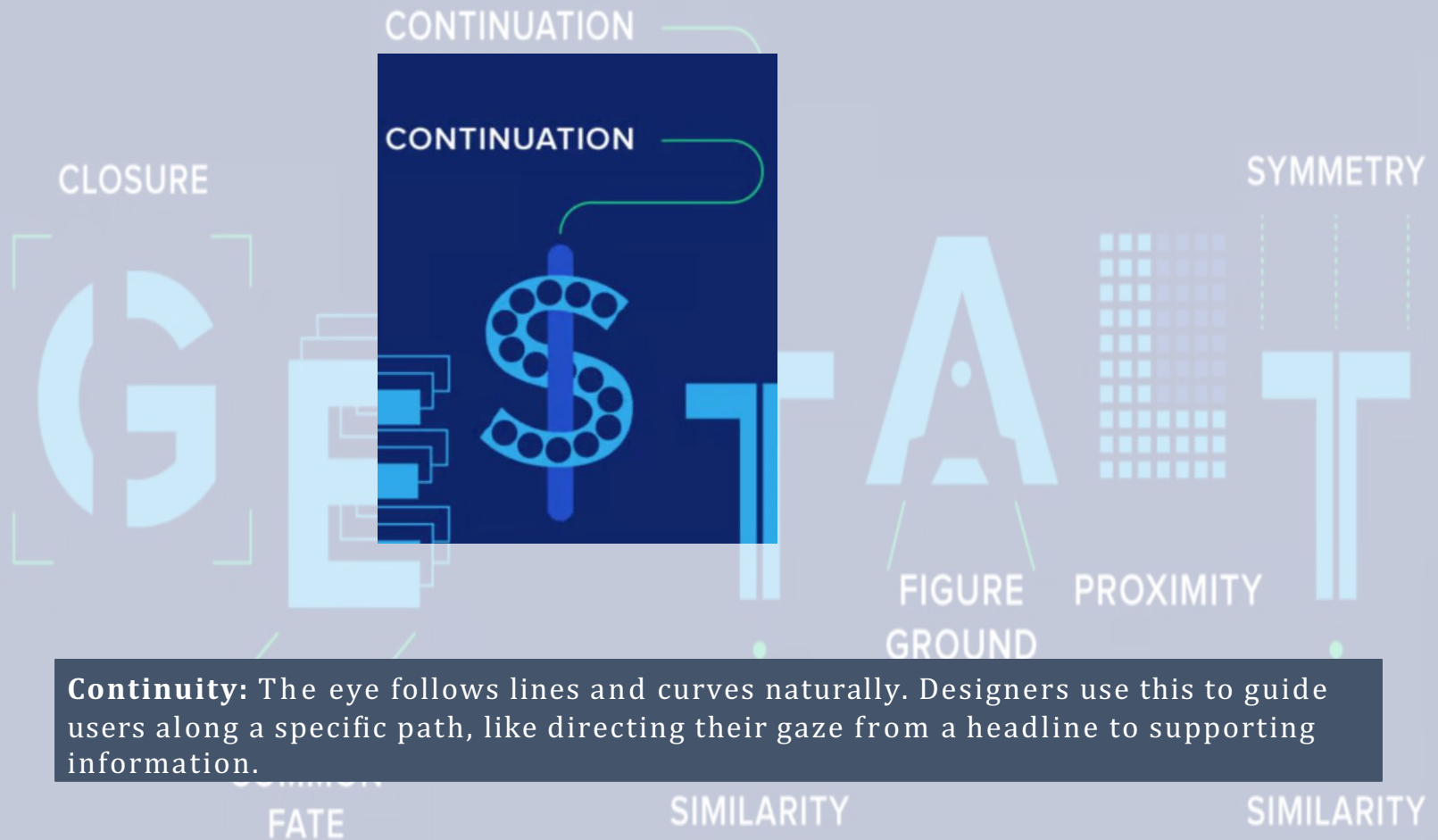
Interaction Design Foundation
interaction-design.org

- This principle refers to the human tendency to perceive visual elements moving in the same direction or in unison as grouped.
- Visuals need not be moving to convey motion.
- Cues such as arrows and the rotation angle can indicate the direction in which the elements are perceived to move.

Common Fate Example



The "Frequently Asked Questions" section on websites is often an accordion. We interpret all the questions as part of a group "moving" in the same direction. In this case, the downward arrows point to the direction each of them will open.

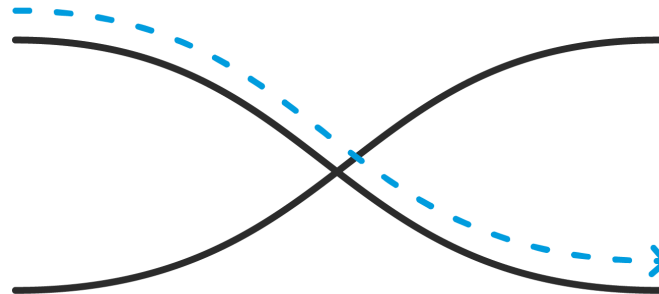


Continuity: The eye follows lines and curves naturally. Designers use this to guide users along a specific path, like directing their gaze from a headline to supporting information.

Image from <https://www.toptal.com/designers/ui/gestalt-principles-of-design>. All rights reserved.

Continuation

Gestalt Rule: **Continuity**



Interaction Design Foundation
interaction-design.org

- The continuity principle of Gestalt states that we group elements that seem to follow a continuous path in a particular direction.
- The human eye follows the paths, lines, and curves of a design and prefers to see a continuous flow of visual elements rather than separated objects.
- The human eye continues to follow the path even if an obstacle hides it or its flow is "broken" by interlinking or bisecting visual elements.

CONTINUATION

Similarity: Similar objects—by color, shape, or size—are perceived as a group. This is useful for establishing consistency and visual relationships in a design.

CLOSURE



COMMON FATE



SYMMETRY

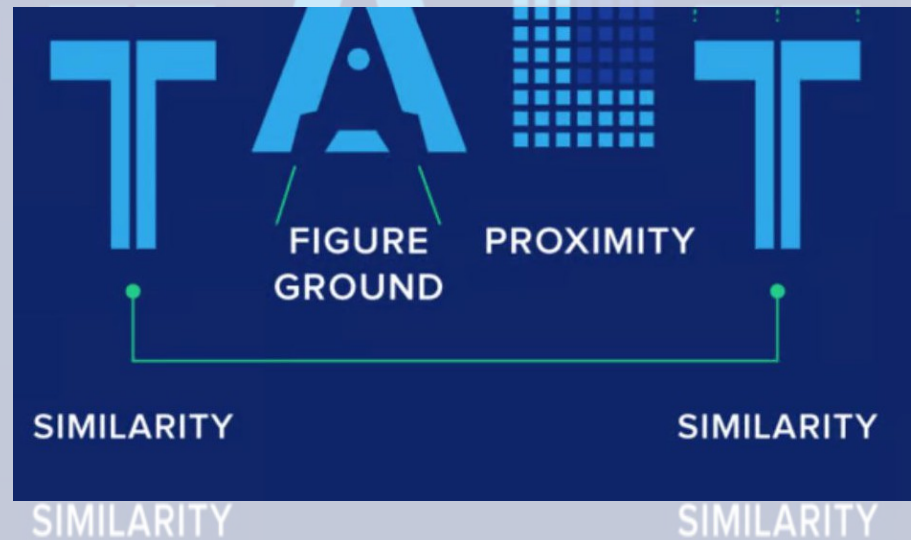
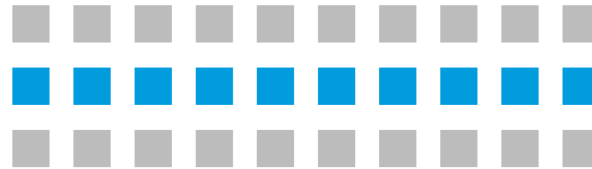


Image from <https://www.toptal.com/designers/ui/gestalt-principles-of-design>. All rights reserved.

Similarity

Gestalt Rule: **Similarity**



Interaction Design Foundation
interaction-design.org

- When items, objects or elements share superficial characteristics, we perceive them as grouped.
- We can see the similarity principle in branding and design system guidelines.

CONTINUATION

Figure-Ground: This principle refers to distinguishing an object from its background. It's essential in creating focus areas, like in imagery where either the figure or ground can take precedence.

RY

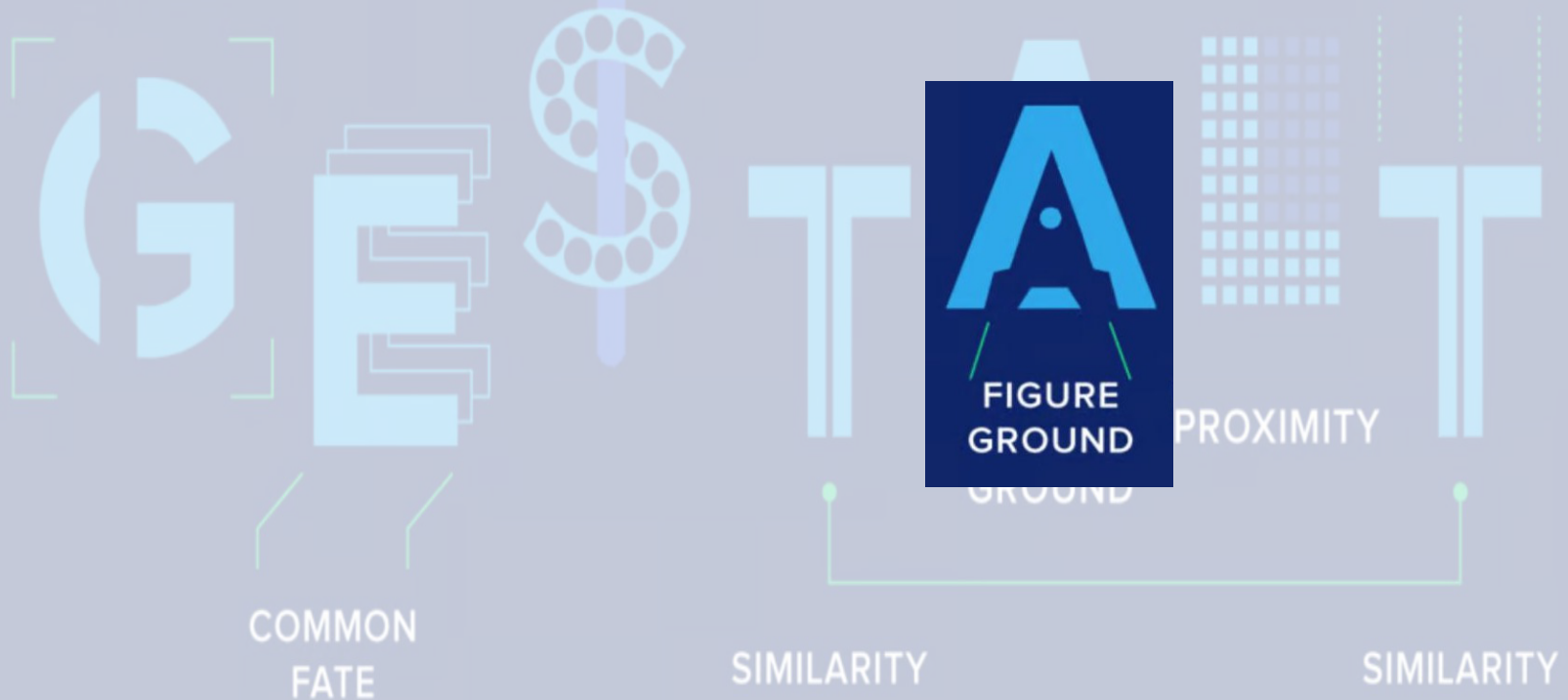


Image from <https://www.toptal.com/designers/ui/gestalt-principles-of-design>. All rights reserved.

Figure/Ground

Gestalt Rule: **Figure/Ground**



Rubin's Vase

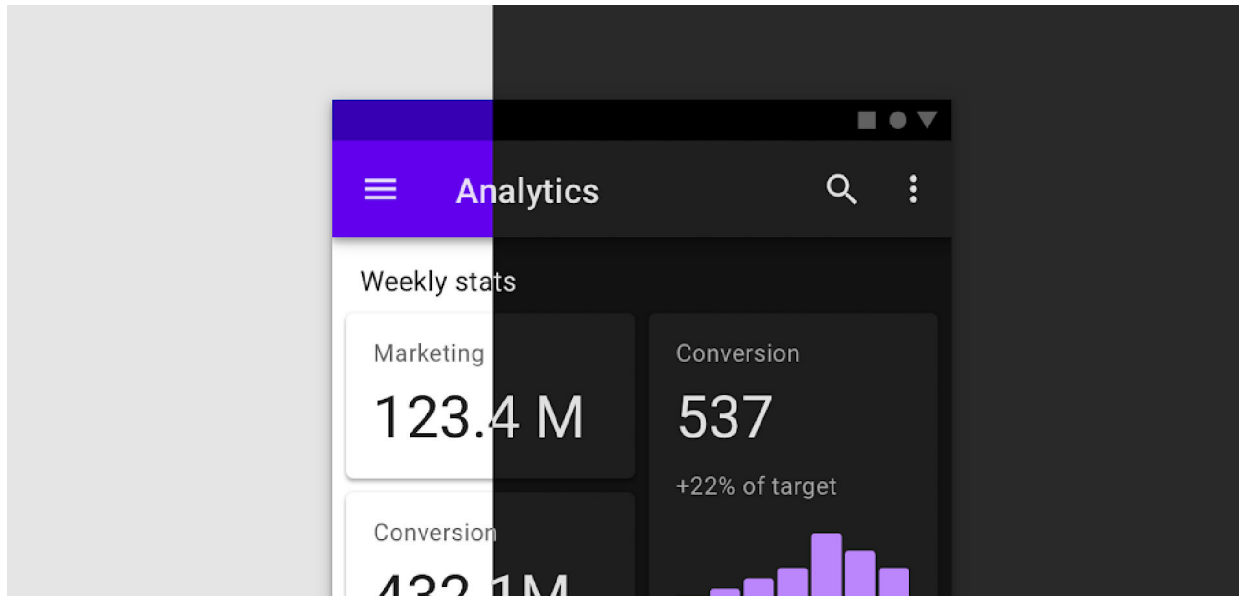
Interaction Design Foundation
interaction-design.org

- We dislike uncertainty, so we look for solid, stable items.
- Unless an image is ambiguous—like Rubin's Vase above—we see its *foreground* first.
- You can apply figure/ground in many ways, but chiefly to contrast elements: for example, light text (i.e., figure) from a dark background (i.e., ground).
- When you use figure/ground well, alongside other considerations such as color theory, you'll help guide users in their tasks and lessen their cognitive load.

Figure/Ground

- Figure/ground and multistability are sometimes confused to be the same.
- However, there is a slight difference.
- In most cases, background and foreground are stable, but in some cases, such as the optical illusion of Rubin's vase, it can contribute to multistability.

Figure/Ground



When an interface's color theme changes from light to dark, the previously black text becomes white, and the white background becomes black. Even though the colors have reversed, we have no trouble recognizing the interface. We automatically interpret the foreground and background colors.

CONTINUATION

Proximity: Elements that are close together are perceived as related or grouped. Designers use this to organize information, like grouping menu items together.

CLOSURE



COMMON FATE



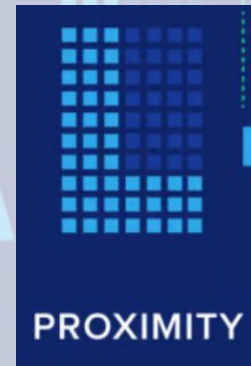
SIMILARITY



FIGURE GROUND



SYMMETRY



SIMILARITY



Image from <https://www.toptal.com/designers/ui/gestalt-principles-of-design>. All rights reserved.

Proximity

Gestalt Rule: **Proximity**



Interaction Design Foundation
interaction-design.org

- We group closer-together elements, separating them from those farther apart.
- When you group elements in your design, users will see it as one distinct entity on the screen.

Proximity



An example of proximity in design is the Girl Scouts logo, with its three faces clustered in profile (two green, one white).

CONTINUATION

Symmetry: Symmetrical elements are seen as part of the same group or whole, providing a sense of balance and harmony. Symmetry in layouts often feels stable and visually pleasing, while asymmetry can add a sense of dynamism.

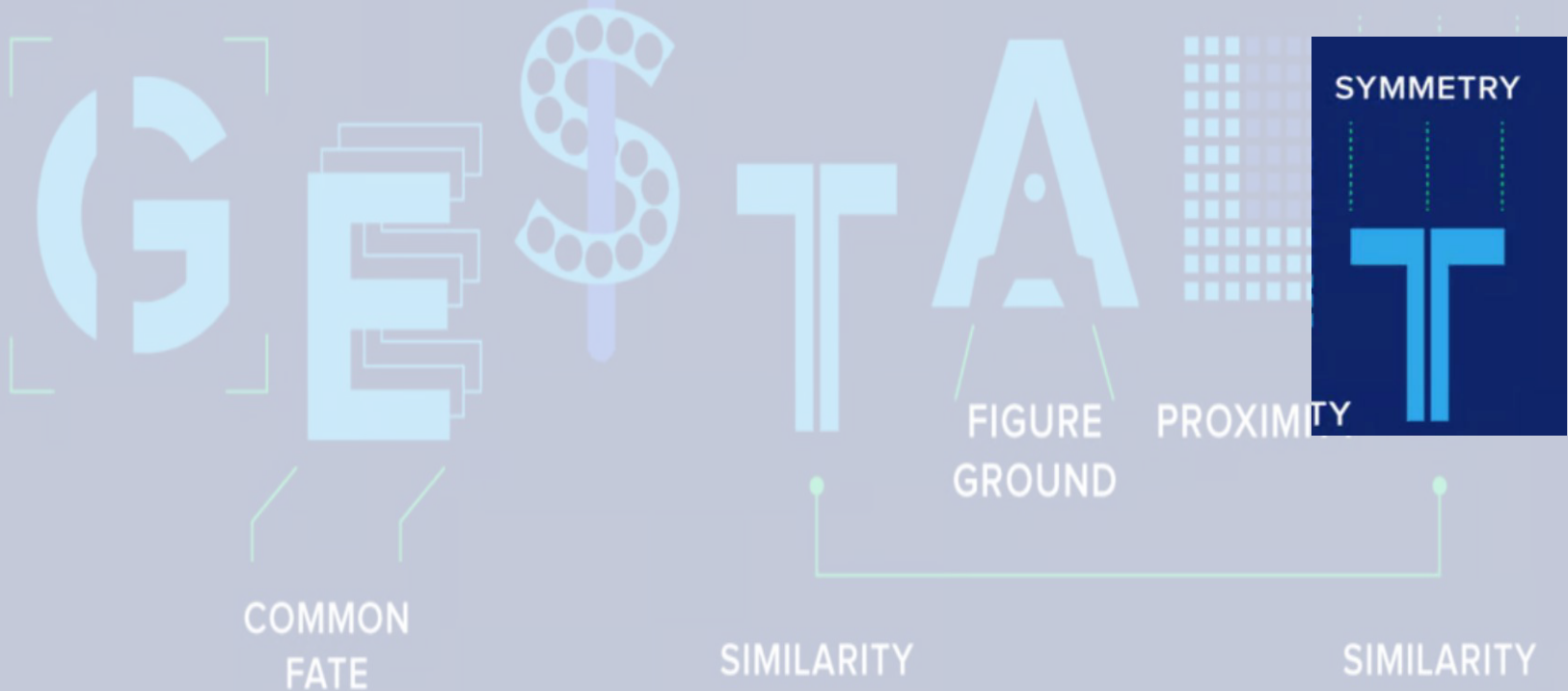
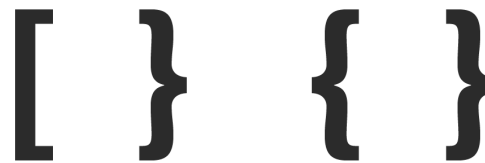


Image from <https://www.toptal.com/designers/ui/gestalt-principles-of-design>. All rights reserved.

Symmetry

Gestalt Rule: **Symmetry**



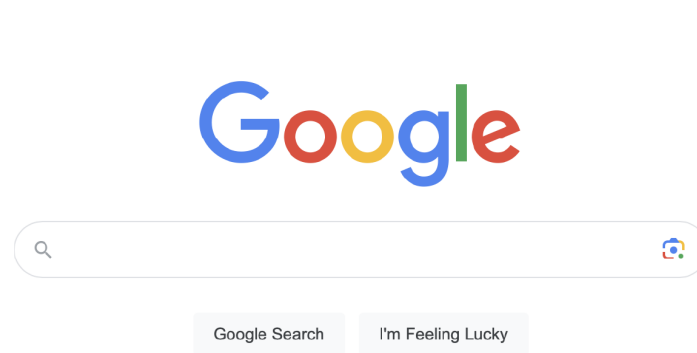
Interaction Design Foundation
interaction-design.org

Which of these shapes would you group together? Chances are, you'll pick the matching curly brackets instead of the mismatched curly and square bracket combination.

Symmetry

- Humans tend to see visual elements as grouped when they are arranged symmetrically.
- The natural world is filled with symmetry (or near symmetry), and our brains tend to favor symmetrical forms. Grid systems that evenly divide the space help designers implement symmetry and order in user interfaces.

Symmetry



Google's home page is symmetrical, with almost all major elements center-aligned and the two buttons, "Google Search" and "I'm Feeling Lucky," nearly mirroring each other.

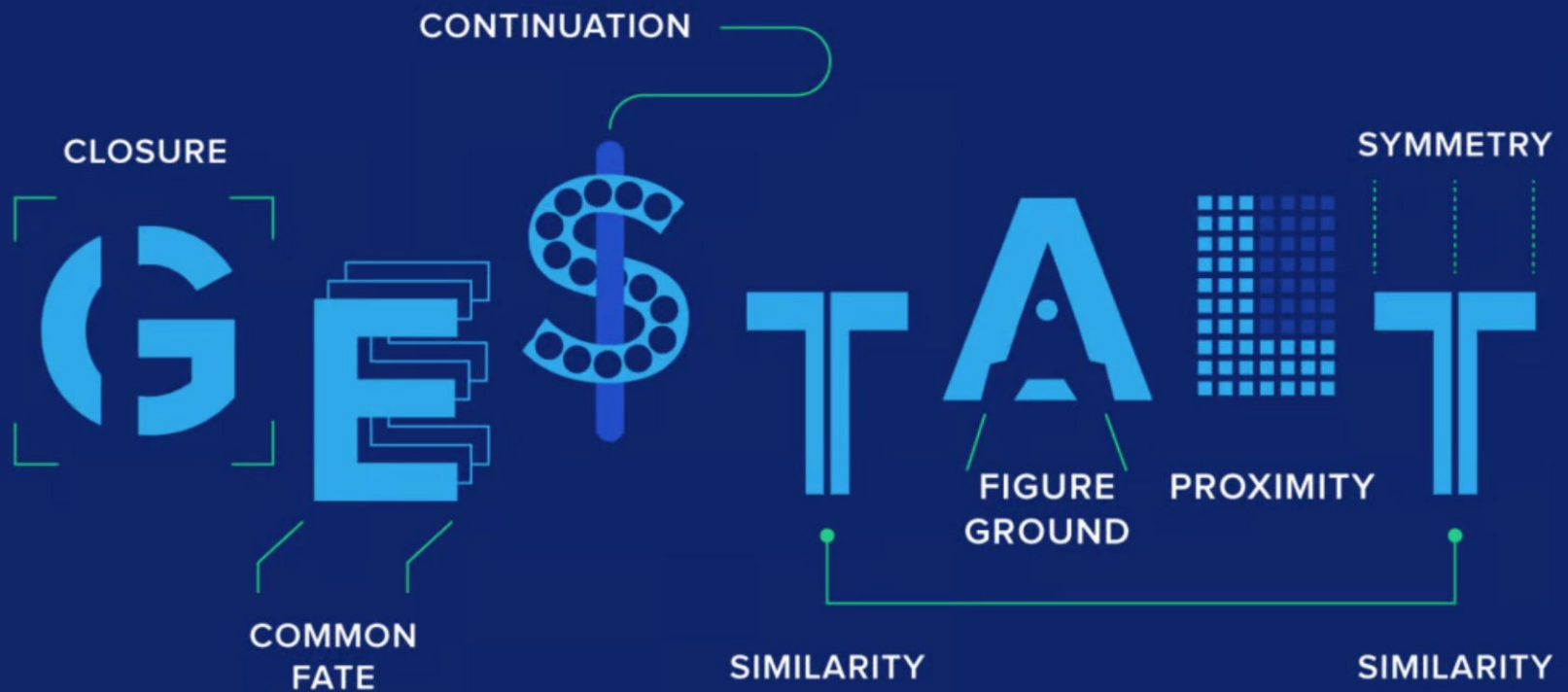


Image from <https://www.toptal.com/designers/ui/gestalt-principles-of-design>. All rights reserved.

Emergence

- We perceive the world without thinking too much about understanding every small thing around us.
- This ability to quickly make sense of our environment is essential for survival.

Gestalt Rule: **Emergence**



Interaction Design Foundation
interaction-design.org



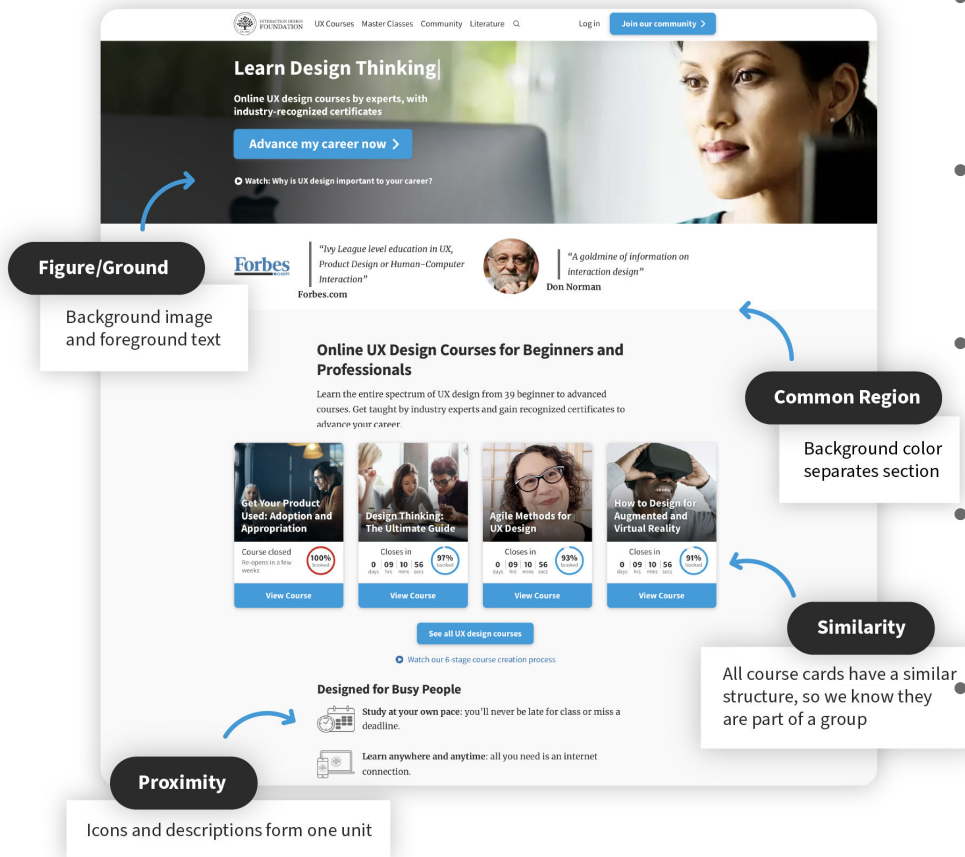
Example



Unilever's logo is composed of several smaller shapes. But the letter "U" emerges from the combination of those smaller elements. Looking further, we see many smaller icons emerge from these abstract shapes.

Gestalt Principles are in the Mind, Not the Eye

- The Gestalt Principles are vital in user experience (UX) design.
- When you design interfaces, users must be able to understand what they see—and find what they want—at a glance.
- Next we will see some examples of the Gestalt principles from the IxDF landing page.



- *The background image and the text overlaid on it demonstrate the principle of figure/ground.*
- *The course cards have a similar structure, so users know they are part of a group.*
- *The icons and descriptions are placed in close proximity to indicate that they belong together.*
- *And finally, colors and graphics divide the page into separate regions.*
- *Without this, users would struggle to make associations between unrelated clustered-together items and leave the site.*

In your designs, you should **never confuse or delay users**.
Instead, **guide them to their options so they can identify
with organizations/brands rapidly.**

It's not just about making things look nice.
How can Gestalt Principles help with
Layout?

Applying Gestalt Principles to Layout

- Hierarchy
- Alignment and Grids
- Whitespace (Negative Space)
- Balance (Symmetry and Asymmetry)
- Consistency and Repetition
- Use of Visual Flow and Continuity

Applying Gestalt Principles to Layout

- **Hierarchy**
 - Establishes clear order of importance
 - Helps guide users on where to focus their attention first
 - Often achieved through contrast, size, and placement.
 - E.g. Homepage: Use of font size and weight, establish a hierarchy between title/logo, menu items, subheadings and body text
 - **Gestalt:** e.g. *continuity* and *similarity* support hierarchy by linking related elements visually, making the flow between elements seamless
- Alignment and Grids
- Whitespace (Negative Space)
- Balance (Symmetry and Asymmetry)
- Consistency and Repetition
- Use of Visual Flow and Continuity

Applying Gestalt Principles to Layout

- Hierarchy
- **Alignment and Grids**
 - Elements look organised and it feels balanced
 - Consistent alignment makes content is easier to scan, and the structure feels more intuitive
 - Grids provide a structured layout through invisible lines which helps with alignment
 - e.g. aligning text boxes, images, buttons, or other design elements
 - **Gestalt:** Grouping elements through *proximity* and using *symmetry* within columns and rows creates a sense of order, make design feel stable
- Whitespace (Negative Space)
- Balance (Symmetry and Asymmetry)
- Consistency and Repetition
- Use of Visual Flow and Continuity

Applying Gestalt Principles to Layout

- Hierarchy
- Alignment and Grids
- **Whitespace (Negative Space)**
 - Unmarked space around elements in a design
 - Purpose:
 - Prevents overcrowding
 - Allows elements to stand out
 - Gives the viewer's eye a place to rest.
 - Sufficient whitespace make a design easier to navigate and less overwhelming
 - e.g. whitespace around images, headings, and paragraphs makes each section clearer and more readable
 - **Gestalt:** Primary element stand out through *Figure-ground* and enough space around an element leads to *Closure*
- Balance (Symmetry and Asymmetry)
- Consistency and Repetition
- Use of Visual Flow and Continuity

Applying Gestalt Principles to Layout

- Hierarchy
- Alignment and Grids
- Whitespace (Negative Space)
- **Balance (Symmetry and Asymmetry)**
 - Arranging elements so that the design feels either stable (symmetrical) or dynamic (asymmetrical).
 - **Symmetrical layouts:**
 - visually balanced
 - feel harmonic and orderly
 - Often used in corporate websites or formal publications
 - **Asymmetrical layouts:**
 - introduce movement and interest
 - guide the eye in specific directions
 - feel more modern or dynamic
 - Often used for a portfolio website to draw attention to a standout piece of work
 - Gestalt: *Symmetry*
- Consistency and Repetition
- Use of Visual Flow and Continuity

Applying Gestalt Principles to Layout

- Hierarchy
- Alignment and Grids
- Whitespace (Negative Space)
- Balance (Symmetry and Asymmetry)
- **Consistency and Repetition**
 - Create familiarity through use of color, typography, and spacing
 - Visual language through repetition of design elements
 - Strongly related to Nielsen's Consistency and Standards
 - **Gestalt: Similarity:** repeating similar elements creates a predictable experience. Similar looking items behave similarly
- Use of Visual Flow and Continuity

Applying Gestalt Principles to Layout

- Hierarchy
- Alignment and Grids
- Whitespace (Negative Space)
- Balance (Symmetry and Asymmetry)
- Consistency and Repetition
- **Use of Visual Flow and Continuity**
 - Visual flow: Guide the user's eye naturally
 - Continuity: Creates a path to follow without abrupt stops
 - E.g.: In a landing page, a clear visual flow leads the user from a headline to a call-to-action button and then to additional information below. Accomplished through visual elements like lines, curves, or directional elements like arrows
 - **Gestalt:** *Continuity* creates a smooth flow reduce cognitive load for the user. *Common fate* directs attention

GOMS

Question?

- How much time will it take to replace all instances of a 4-letter word in MS Word?

What is GOMS?

- A family of user interface modeling techniques
- Goals, Operators, Methods, and Selection rules
 - Input: detailed description of UI and task(s)
 - Output: various qualitative and quantitative measures

What can GOMS model?

- Task must be goal-directed
 - Not problem solving or creative tasks (e.g. design)
- Can include serial and parallel tasks

How to Do GOMS Analysis

- Generate task description
 - Pick high-level user Goal
 - Write Method for accomplishing Goal - may invoke subgoals
 - Write Methods for subgoals
 - This is recursive
 - Stops when Operators are reached
- Evaluate description of task
- Apply results to UI

Operators vs. Methods

- Operator: the most primitive action
- Method: requires several Operators or subgoal invocations to accomplish

An Example: Keystroke-Level Model

- How to make a KLM
 - List specific actions user does to perform task
 - Keystrokes and button presses
 - Mouse movements
 - Hand movements between keyboard & mouse
 - System response time (if it makes user wait)
 - Add Mental operators
 - Assign execution times to steps
 - Add up execution times
- Only provides execution time and operator sequence

KLM Example

- Replace all instances of a 4-letter word.

(example from Hochstein)

Description	Operation	Time (sec)
Reach for mouse	H[mouse]	0.40
Move pointer to "Replace" button	P[menu item]	1.10
Click on "Replace" command	K[mouse]	0.20
Home on keyboard	H[keyboard]	0.40
Specify word to be replaced	M4K[word]	2.15
Reach for mouse	H[mouse]	0.40
Point to correct field	P[field]	1.10
Click on field	K[mouse]	0.20
Home on keyboard	H[keyboard]	0.40
Type new word	M4K[word]	2.15
Reach for mouse	H[mouse]	0.40
Move pointer on Replace-all	P[replace-all]	1.10
Click on field	K[mouse]	0.20
Total		10.2

According to this KLM model, it takes 10.2 seconds to accomplish this task.

Advantages of GOMS

- Gives several qualitative and quantitative measures
- Model explains **why** the results are what they are
- Less work than usability study
- Easy to modify when interface is revised
- Research ongoing for tools to aid modeling process

Disadvantages of GOMS

- Only works for goal-directed tasks
- Assumes tasks are performed by expert users
- Evaluator must pick users' tasks/goals
- Does not address several important UI issues, such as
 - readability of text
 - memorability of icons, commands

Questions?